

Composite Silhouette

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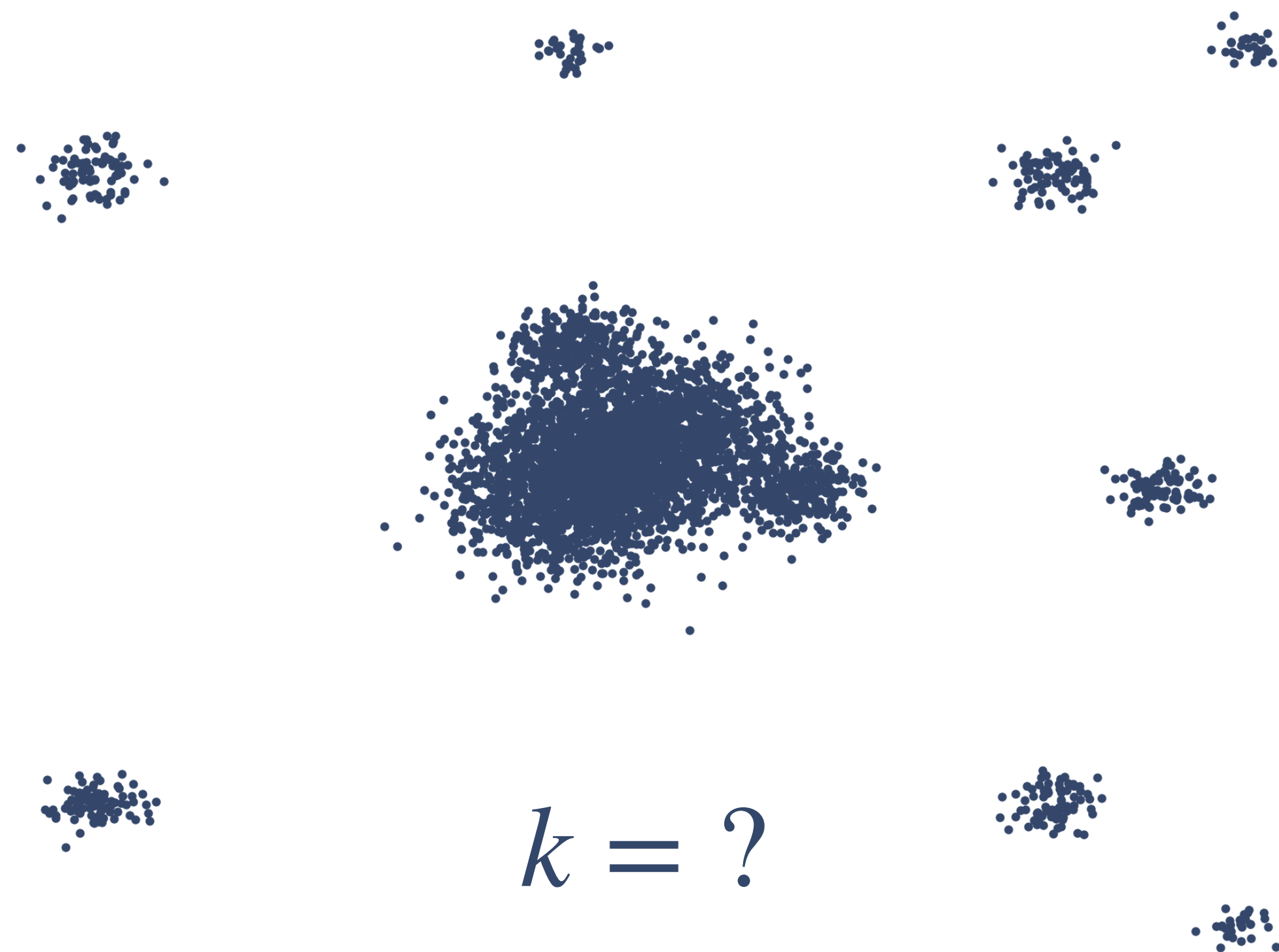
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How many clusters are there?



Selection Approaches

Heuristic / Statistical Approaches

Elbow Method: *Bend in within-cluster variation*

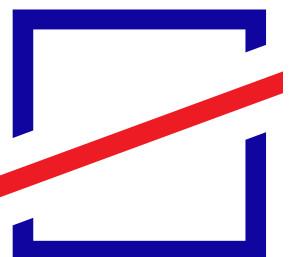
Gap Statistic: *Comparison with a reference distribution*

Internal Validation Indices

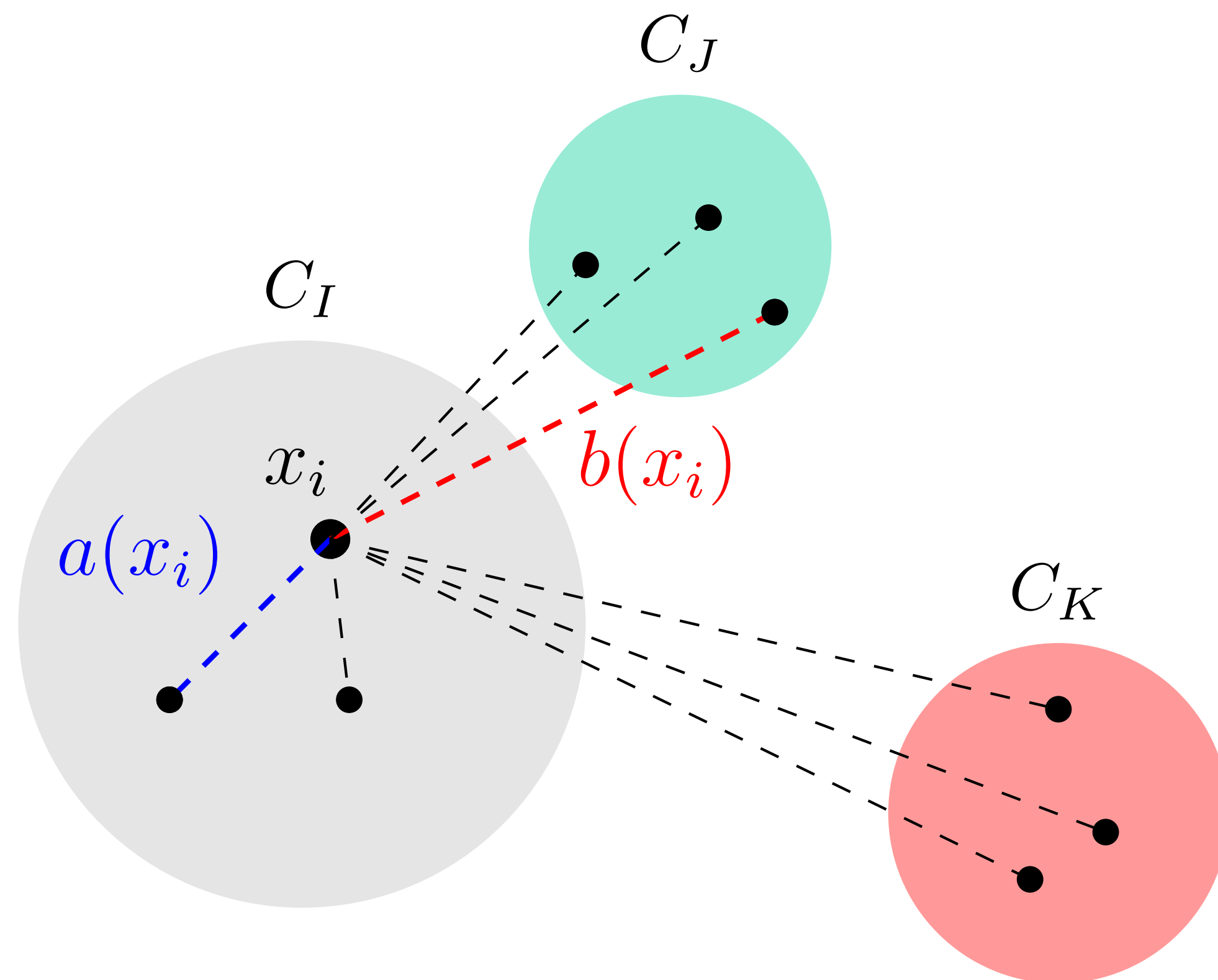
Calinski-Harabasz: *Between- vs. within-cluster dispersion*

Davies-Bouldin: *Cluster scatter vs. separation*

Silhouette: *Cohesion vs. separation*



Silhouette Coefficient



$a(x_i)$ = Average distance to points in its own cluster

$b(x_i)$ = Minimum average distance to another cluster

Individual Silhouette

$$s(x_i) = \frac{b(x_i) - a(x_i)}{\max\{a(x_i), b(x_i)\}}, \quad -1 \leq s(x_i) \leq 1$$

Higher values indicate that an observation is well matched to its own cluster and well separated from neighboring clusters.

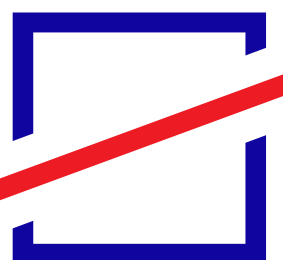
Aggregation $s(x_1), s(x_2), \dots, s(x_n) \rightarrow S$

How should the Silhouette values be aggregated?

Two common aggregation strategies

Micro Silhouette (S_m): Average across all data points

Macro Silhouette (S_M): Average across clusters



Silhouette Coefficient: Micro vs. Macro

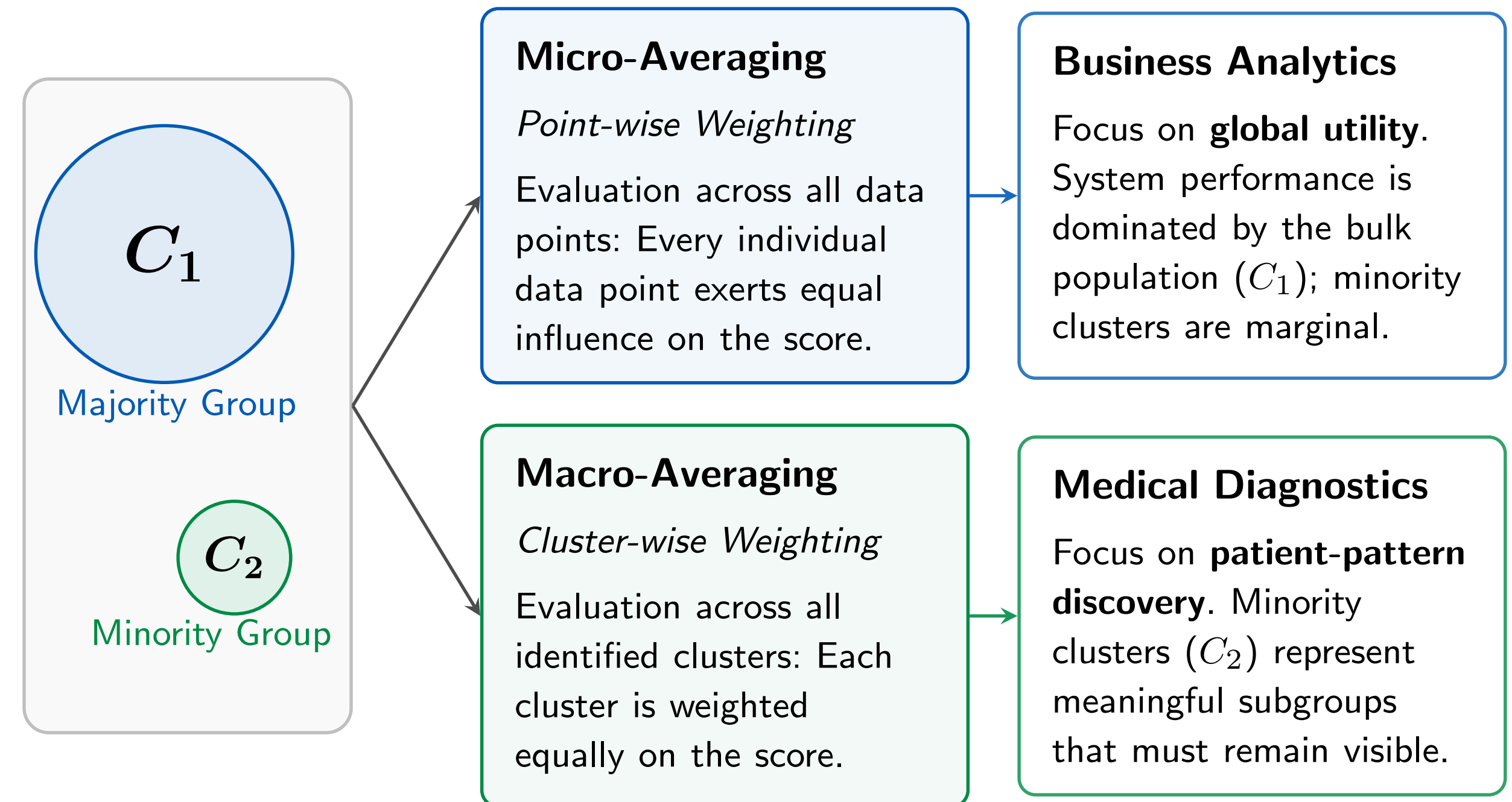
Different weighting, different behavior

Micro and Macro averaging emphasise different aspects of the same partition.

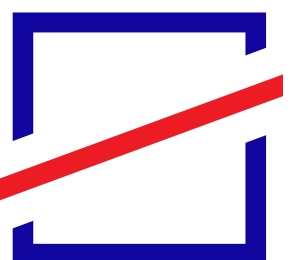
Micro-averaging can exhibit a bias toward larger clusters, potentially masking poor clustering quality in smaller ones.

Macro-averaging mitigates this bias by weighting clusters equally, but may instead overemphasise small or under-represented clusters.

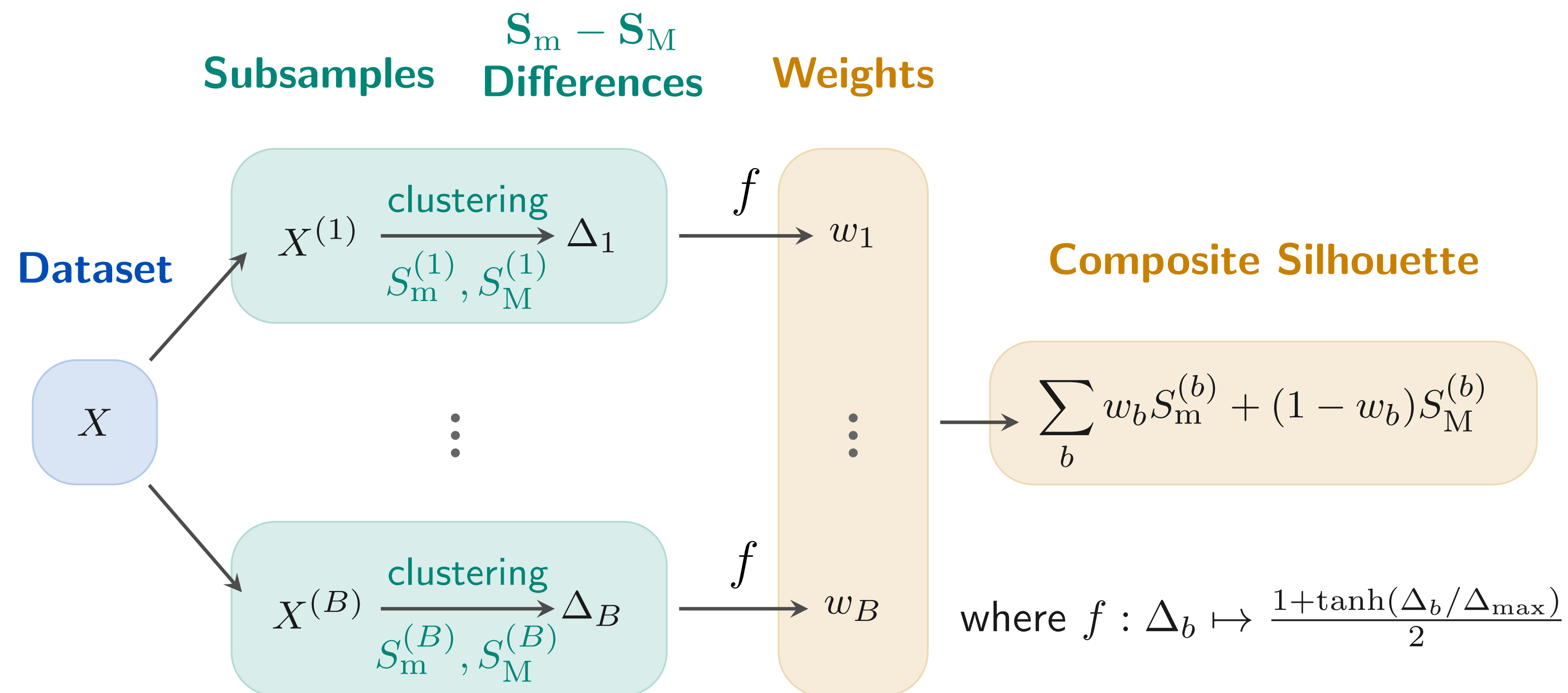
Neither strategy is universally preferable. The appropriate choice may depend on the application, while in many settings it is unclear whether majority structure or minority representation should receive greater emphasis.



Can we combine the strengths of Micro and Macro Silhouette to select the number of clusters more reliably, without assuming which aggregation strategy is preferable?



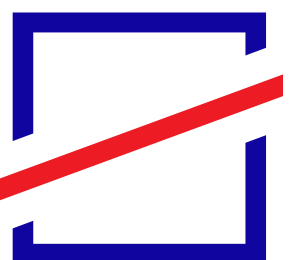
Composite Silhouette



Composite Silhouette considers multiple subsampled views of the data, each of which is clustered and evaluated using both Micro S_m and Macro Silhouette S_M .

In each subsampled clustering, the difference between the two scores is computed and scaled by the maximum observed difference across subsamples. A nonlinear transformation then maps this scaled difference into an adaptive weight, determining the relative contribution of Micro and Macro Silhouette.

Finally, the weighted scores are averaged across subsamples to obtain the Composite Silhouette score.





Composite Silhouette

Repeated views of the structure

Subsampling provides multiple views of the data, allowing the relationship between Micro- and Macro-averaged Silhouette to be observed across different clustering solutions.

Disagreement as local information

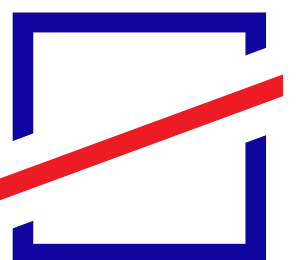
When one aggregation gives a stronger Silhouette value on a subsample, the composite shifts towards that perspective. The magnitude of the disagreement determines how strong this shift is, while the other perspective is never discarded.

Beyond a single clustering solution

Aggregating across subsamples captures how the relative support for Micro and Macro varies across different views of the data, providing a broader assessment of the underlying clustering structure.

Cluster-count selection

Given a clustering algorithm, Composite Silhouette is evaluated across candidate numbers of clusters, and the cluster count with the highest score is selected.





Theoretical Analysis

To study the statistical behaviour of Composite Silhouette, we consider an independent estimate of the normalization term, which allows us to apply standard concentration arguments.

Boundedness

Each subsample level Composite Silhouette score is a convex combination of Micro and Macro Silhouette and therefore remains bounded in $[-1, 1]$.

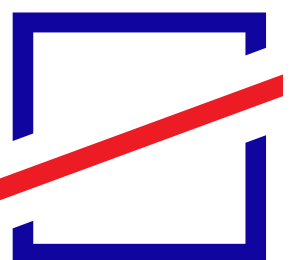
Concentration

Using **Hoeffding's inequality** and a **union bound** over the candidate cluster counts, we obtain a uniform concentration guarantee: as the number of subsamples B increases, the estimated Composite Silhouette curve becomes increasingly stable, with error decreasing at rate $O(B^{-1/2})$.

Recovery

Under a **positive-margin condition**, this concentration translates into increasingly reliable recovery of the optimal candidate number of clusters as the number of subsamples increases.

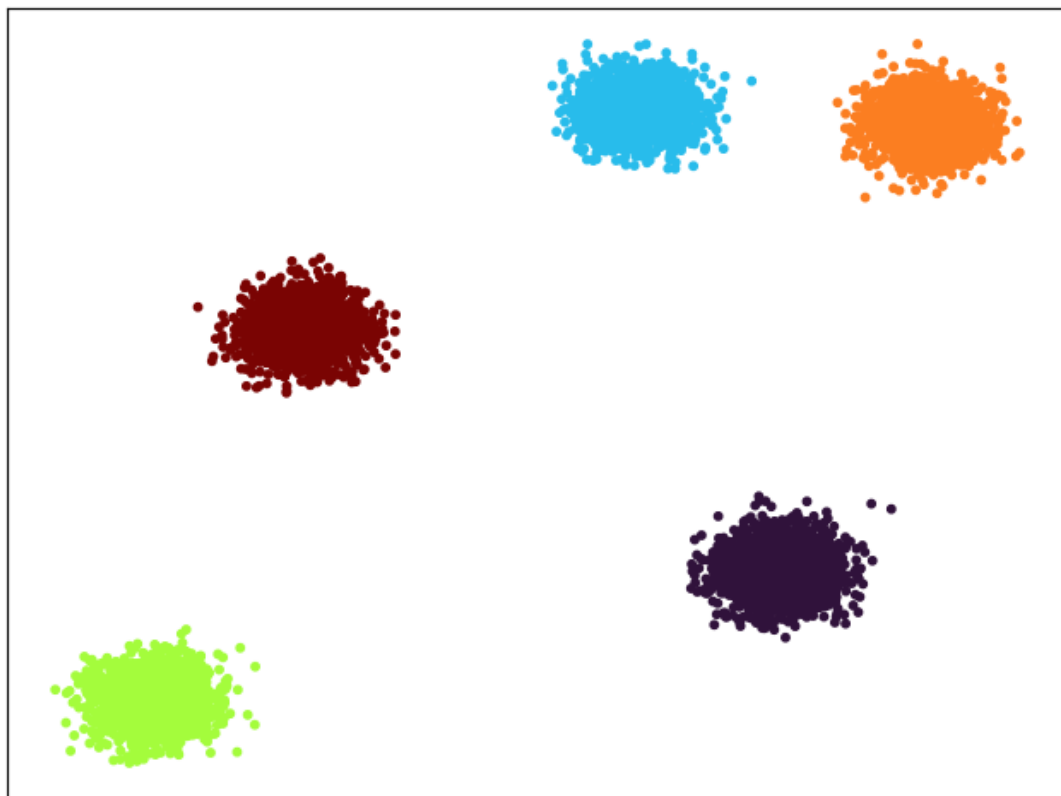
More subsamples → stronger concentration → more reliable cluster-count selection



Experimental Setup

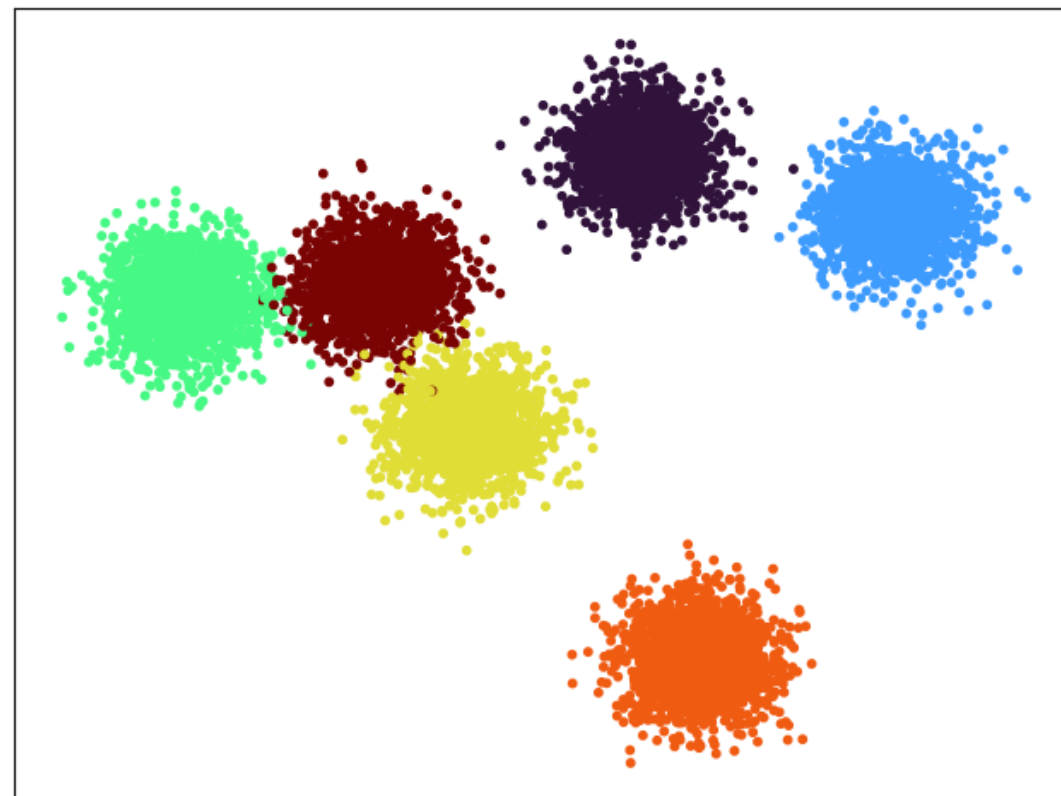
Synthetic datasets: Controlled settings designed to expose different behaviour of Micro and Macro averaging.

S1 - Clean & well-separated



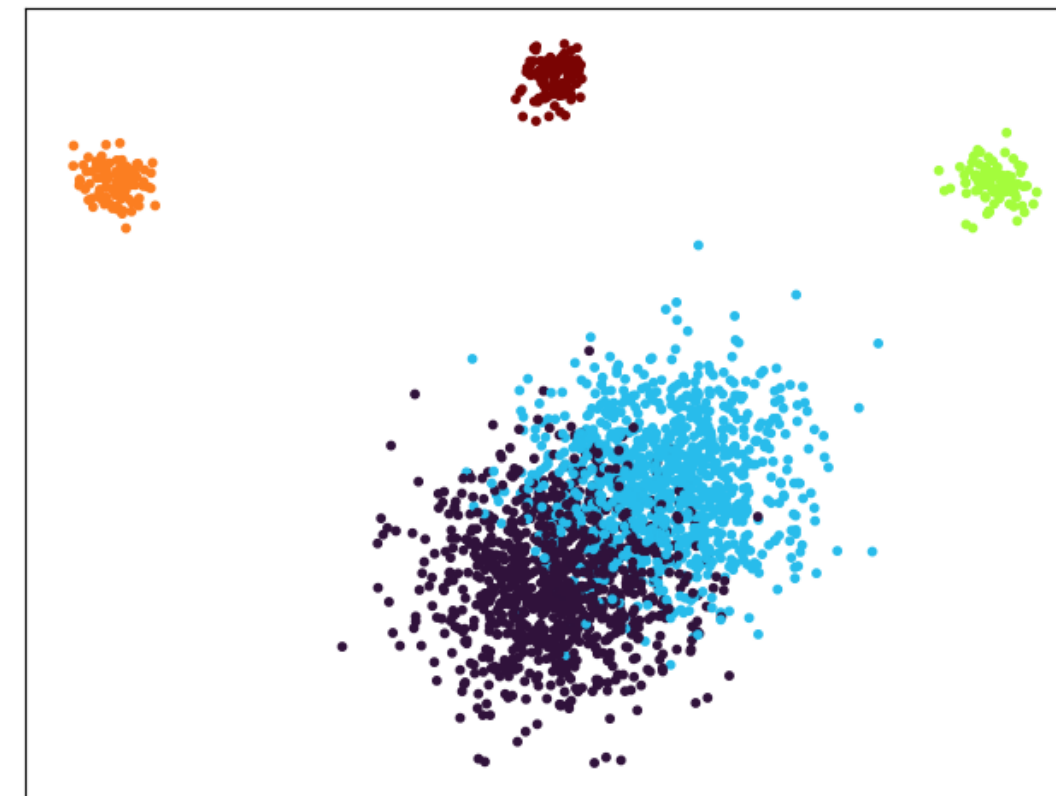
Both strategies expected to be informative.

S2 - Balanced & more diffuse



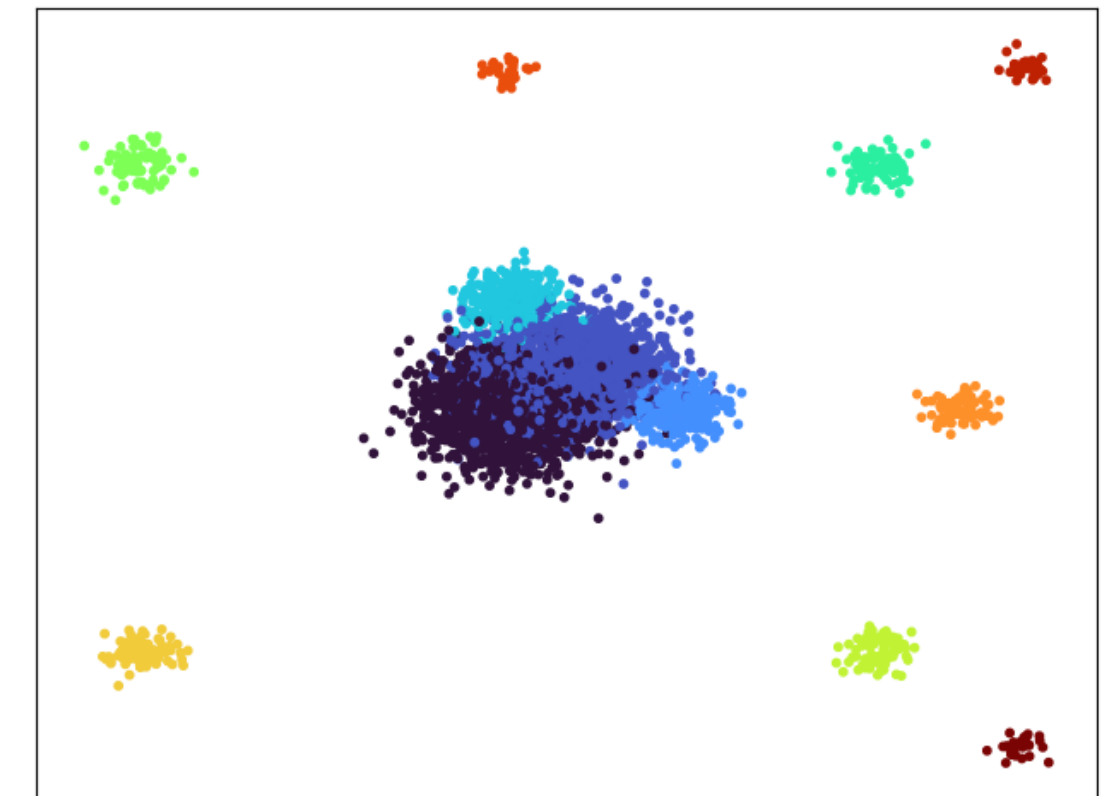
Micro expected to be more informative.

S3 - Size & scale imbalance



Macro expected to be more informative.

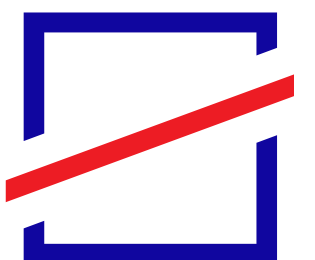
S4 - Strong heterogeneity



Neither strategy expected to be consistently informative.

Real-world datasets: Evaluation across diverse domains, modalities, dataset sizes, and cluster structures.

Parkinsons (Tabular) • **Wine** (Tabular) • **Blood Transfusion** (Tabular) • **Digits** (Vision) • **BBC News** (Text) • **HTRU2** (Tabular),
STL-10 (Vision) • **20 Newsgroups** (Text) • **Spambase** (Tabular) • **Minds14** (Text) • **Bank Marketing** (Tabular) • **Banking77** (Text)





Experimental Setup

Comparison with existing criteria

Composite Silhouette is compared with **Micro and Macro Silhouette obtained from the same subsamples and averaged across them**. We additionally compare with **Micro Silhouette, Macro Silhouette, Calinski-Harabasz, Davies-Bouldin and Elbow**, averaged across repeated clustering runs, as well as the **Gap Statistic**. For a fair comparison, the number of repeated runs matches the number of subsamples used by Composite Silhouette.

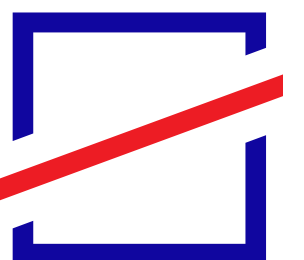
General subsampling strategy

We use 20 subsamples, with the subsample size determined by a predefined general rule based on the dataset size and candidate cluster counts, rather than tuning it separately for each dataset.

Cluster-count recovery

All criteria use the same candidate cluster counts, within ± 5 of the ground-truth number of clusters and starting from 2. Each criterion selects its optimal candidate, allowing us to assess whether it recovers the ground-truth number of clusters.

The evaluation is repeated for each dataset and each clustering algorithm considered: k -means, Gaussian Mixture Models and Bisecting k -means.

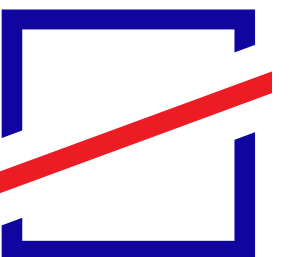


Synthetic Dataset Evaluation

Composite Silhouette recovers the ground-truth cluster count across all synthetic settings. It follows the informative aggregation strategy when one is sufficient, while remaining effective when neither Micro nor Macro succeeds alone.

	S1	S2	S3	S4
Ground-truth number of clusters	5	6	5	12
Elbow Method	5 ✓	3 ✗	5 ✓	9 ✗
Gap Statistic	5 ✓	6 ✓	3 ✗	14 ✗
Micro Silhouette	5 ✓	6 ✓	2 ✗	9 ✗
Macro Silhouette	5 ✓	3 ✗	5 ✓	9 ✗
Calinski-Harabasz	5 ✓	6 ✓	10 ✗	16 ✗
Davies-Bouldin	5 ✓	6 ✓	5 ✓	9 ✗
Subsample-averaged Micro Silhouette	5 ✓	6 ✓	2 ✗	8 ✗
Subsample-averaged Macro Silhouette	5 ✓	3 ✗	5 ✓	13 ✗
Composite Silhouette	5 ✓	6 ✓	5 ✓	12 ✓

Number of clusters selected by each criterion under k -means; similar trends are observed with GMMs and Bisecting k -means.

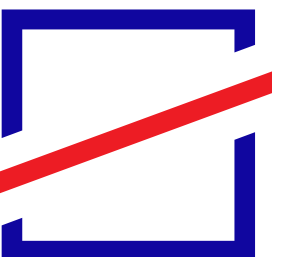


Real-World Dataset Evaluation

Composite Silhouette is the only criterion to recover the ground-truth cluster count across all real-world datasets. When either Micro or Macro provides the informative perspective, Composite follows it adaptively, without requiring a predefined choice between the two. This shows that the adaptive behaviour observed in the controlled synthetic settings also extends to diverse real-world clustering structures.

	Pks	Wne	Bld	Dgt	Bbc	Htr	Stl	Nsg	Spm	Mds	Bnk	B77
Ground-truth number of clusters	2	3	2	10	5	2	10	20	2	14	2	77
Elbow Method	3 ✗	3 ✓	4 ✗	3 ✗	5 ✓	4 ✗	8 ✗	21 ✗	3 ✗	14 ✓	8 ✗	81 ✗
Gap Statistic	7 ✗	3 ✓	5 ✗	10 ✓	9 ✗	7 ✗	14 ✗	24 ✗	3 ✗	16 ✗	2 ✓	75 ✗
Micro Silhouette	2 ✓	3 ✓	2 ✓	10 ✓	5 ✓	2 ✓	6 ✗	23 ✗	2 ✓	15 ✗	2 ✓	81 ✗
Macro Silhouette	2 ✓	3 ✓	6 ✗	9 ✗	5 ✓	2 ✓	6 ✗	24 ✗	2 ✓	14 ✓	2 ✓	77 ✓
Calinski-Harabasz	3 ✗	3 ✓	6 ✗	3 ✗	2 ✗	4 ✗	6 ✗	24 ✗	3 ✗	14 ✓	8 ✗	72 ✗
Davies-Bouldin	2 ✓	3 ✓	4 ✗	9 ✗	7 ✗	3 ✗	10 ✓	24 ✗	2 ✓	14 ✓	8 ✗	81 ✗
Subsample-averaged Micro Silhouette	2 ✓	3 ✓	2 ✓	10 ✓	6 ✗	2 ✓	10 ✓	24 ✗	2 ✓	15 ✗	2 ✓	82 ✗
Subsample-averaged Macro Silhouette	2 ✓	3 ✓	6 ✗	10 ✓	5 ✓	2 ✓	10 ✓	21 ✗	2 ✓	12 ✗	2 ✓	77 ✓
Composite Silhouette	2 ✓	3 ✓	2 ✓	10 ✓	5 ✓	2 ✓	10 ✓	20 ✓	2 ✓	14 ✓	2 ✓	77 ✓

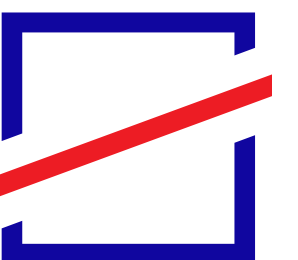
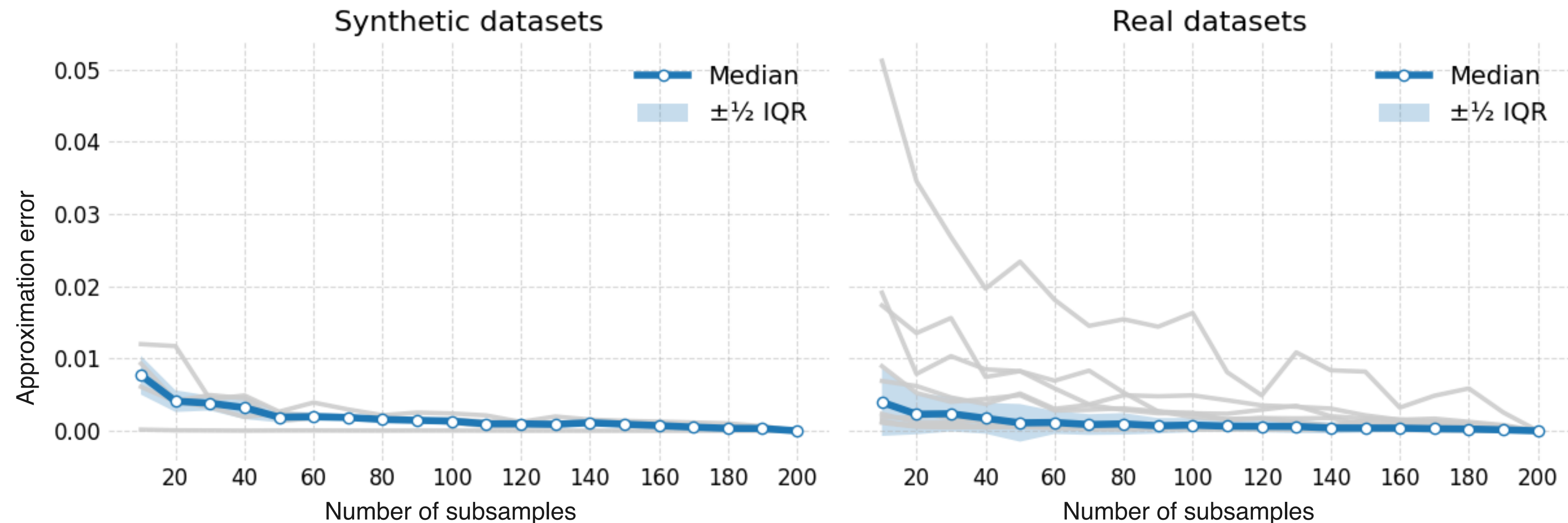
Number of clusters selected by each criterion under k -means; similar trends are observed with GMMs and Bisecting k -means.



Stability across Subsamples

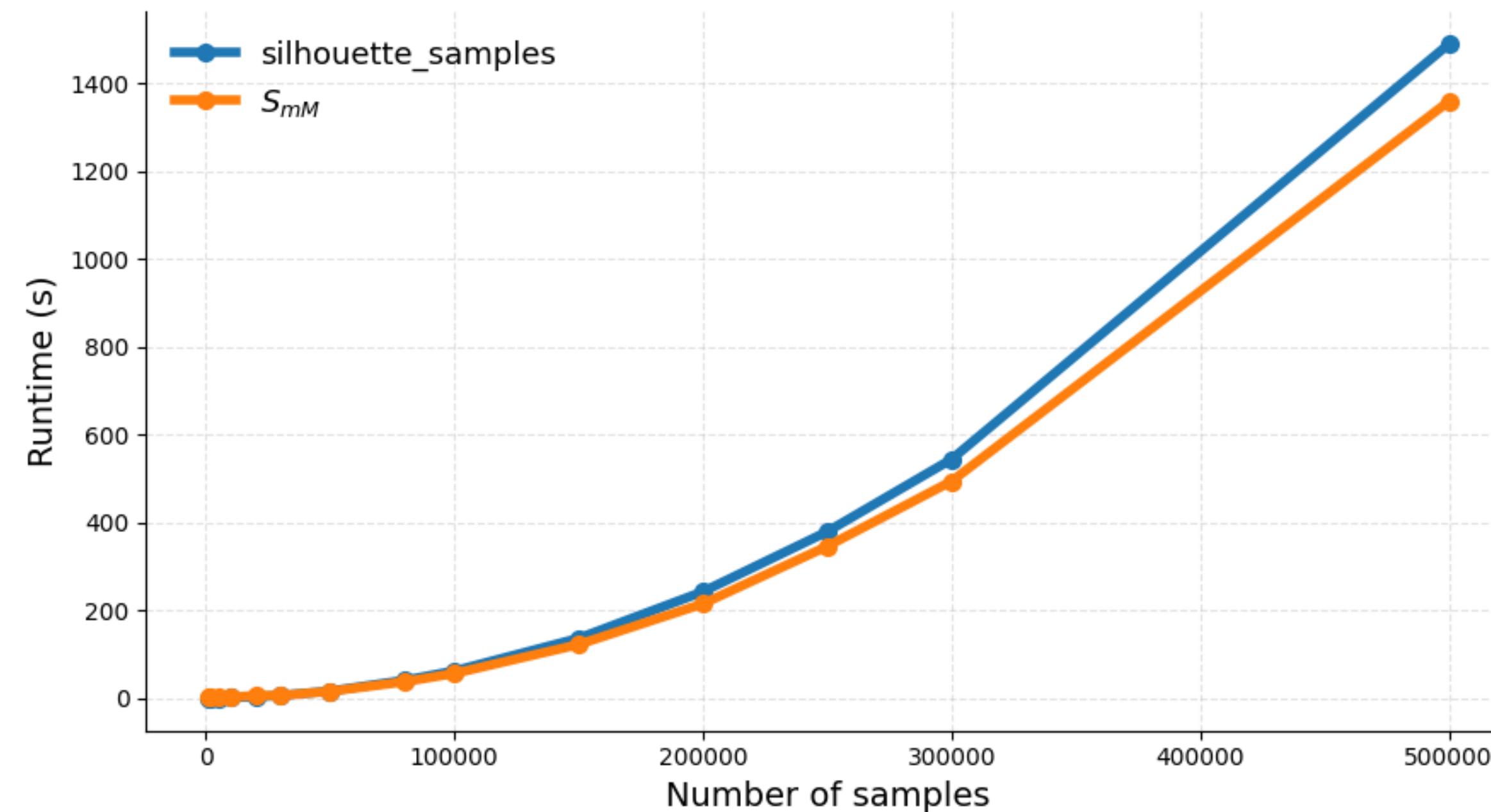
We evaluate the stability of Composite Silhouette by varying the number of subsamples and measuring the approximation error relative to a high-precision estimate obtained using 200 subsamples. The approximation error decreases rapidly as the number of subsamples increases.

Low approximation error is already achieved with a relatively small number of subsamples (median error < 0.01 with only 10 subsamples), with further increases providing progressively smaller improvements.



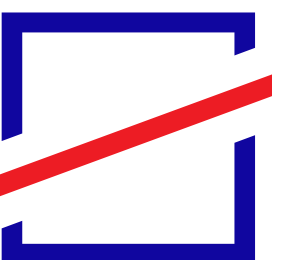
Computational Scalability

We evaluate computational scalability by comparing the cost of computing Composite Silhouette with standard full-data Silhouette as the dataset size increases for a fixed number of clusters. Composite Silhouette remains computationally competitive and requires less runtime as the dataset size grows.



Runtime comparison on a synthetic dataset with five Gaussian clusters, increasing the number of samples from 1,000 to 500,000.

(S_{mM} : Composite Silhouette; `silhouette_samples`: scikit-learn Silhouette implementation)





Conclusions

Adaptive combination of Micro- and Macro-averaged Silhouette

Composite Silhouette combines the two perspectives locally across subsampled clustering, using their disagreement to determine their relative contribution.

Consistent across diverse settings

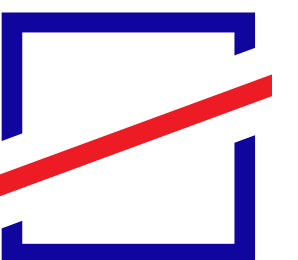
Composite Silhouette was the only method to correctly recover the ground-truth number of clusters across all 16 datasets, including balanced, imbalanced, and heterogeneous structures.

No prior choice between Micro and Macro

It does not require us to decide beforehand which aggregation strategy should be preferred, and remains effective when neither perspective succeeds alone.

Stable and computationally practical

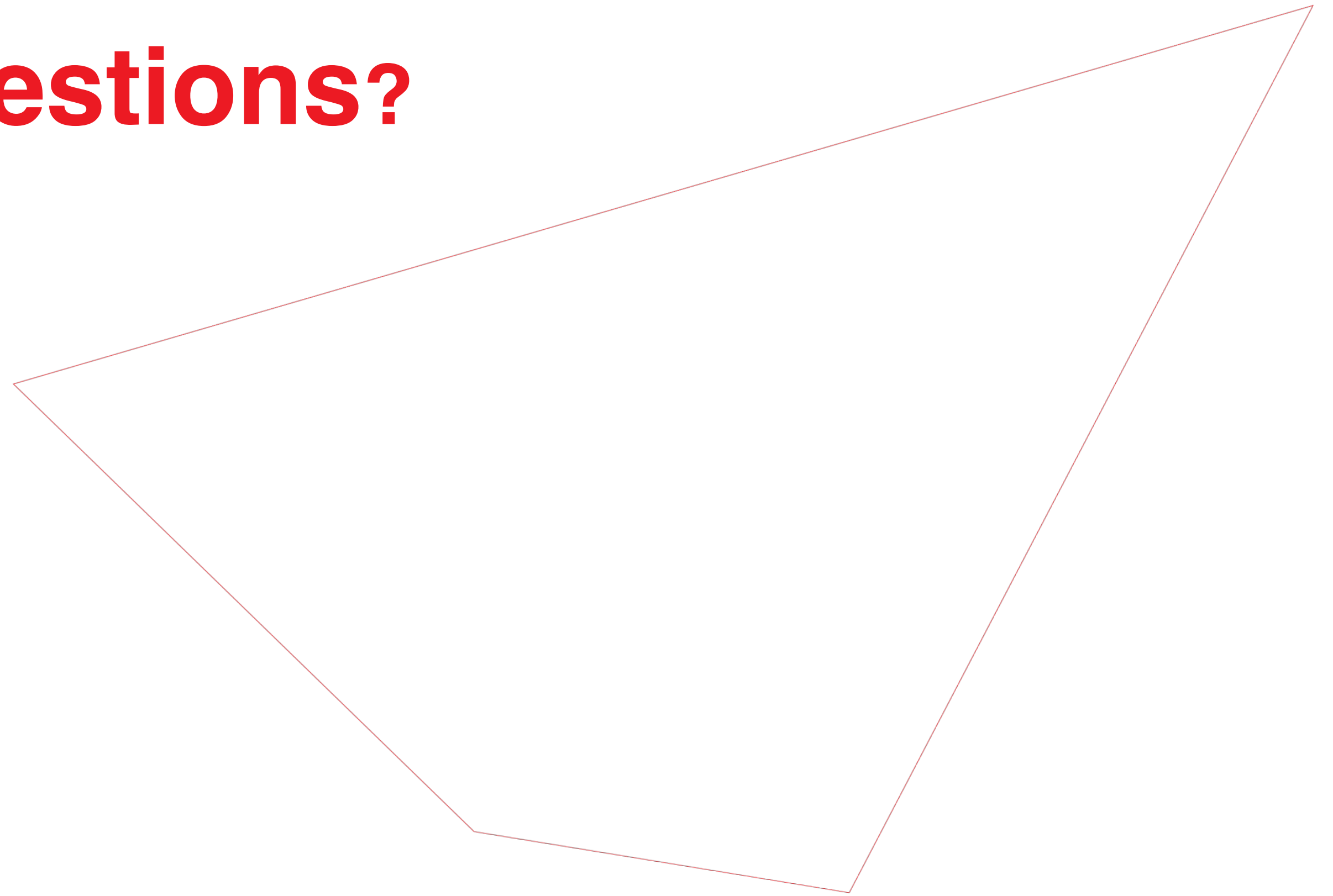
A relatively small number of subsamples is sufficient for stable estimation, while subsampling keeps the approach computationally competitive as dataset size grows.



Thank you!



Questions?



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